NONEXISTENCE OF A CYCLIC $(963, 222, 51)$ DIFFERENCE SET

Abstract. We show that the cyclic group C_{963} admits no $(963, 222, 51)$-difference set. The proof uses the quotient $C_{963} \twoheadrightarrow C_{107}$. A contracted multiplier theorem makes 19 a 107-multiplier, so after translation the quotient coefficient function is constant on the multiplication-by-19 orbits on C_{107}. Those orbits have sizes $1, 53, 53$, while each quotient coefficient is at most 9. The required weight 222 is therefore impossible.

1. Introduction

The cyclic row $(963, 222, 51)$, equivalently the $[9, 107]$ case, appears as a small open residue in Gordon's La Jolla difference-set status tables [2]. We give a short quotient-orbit obstruction that rules it out.

Theorem 1. *The cyclic group C_{963} does not admit a $(963, 222, 51)$-difference set.*

Proof. Assume for contradiction that $D \subseteq G = C_{963}$ is a $(963, 222, 51)$-difference set. Set

$$n = k - \lambda = 222 - 51 = 171 = 3^2 \cdot 19.$$

Let N be the subgroup of order 9, and write

$$Q = G/N \cong C_{107}.$$

For each $u \in Q$, let a_u denote the number of points of D in the coset above u. Then

$$0 \le a_u \le 9 \qquad \text{and} \qquad \sum_{u \in Q} a_u = 222.$$

We use the contracted multiplier theorem on the quotient Q. Take $t = 19$ and $w = 107$. For the prime divisor 19 of n we have $19^1 \equiv 19 \pmod{107}$, and for the prime divisor 3 of n we have

$$3^{95} \equiv 19 \pmod{107}.$$

Therefore Baumert–Gordon's sufficient criterion for w-multipliers applies, so 19 is a 107-multiplier for D [1, Thm. 3.2]. Equivalently, after translating D if necessary, we may assume that the quotient coefficient function is invariant under multiplication by 19 on Q:

$$a_{19u} = a_u \qquad (u \in Q).$$

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Since 107 is prime, the orbit sizes of $u \mapsto 19u$ are determined by $\mathrm{ord}_{107}(19)$. A direct modular computation gives

$$\mathrm{ord}_{107}(19) = 53.$$

Hence the action of multiplication by 19 on Q has exactly three orbits:

$$\{0\}, \qquad O_1, \qquad O_2,$$

where $|O_1| = |O_2| = 53$. Because the function $u \mapsto a_u$ is orbit-constant, there exist integers b_0, α, β with

$$0 \leq b_0, \alpha, \beta \leq 9$$

such that

$$222 = b_0 + 53(\alpha + \beta).$$

Reducing modulo 53 gives

$$b_0 \equiv 222 \equiv 10 \pmod{53}.$$

But $0 \leq b_0 \leq 9$, which is impossible. This contradiction proves that C_{963} admits no $(963, 222, 51)$-difference set. □

Remark 2. The decisive obstruction takes place on the prime quotient C_{107}. The weight 222 is too small in the distinguished fixed coset and too rigid modulo the two 53-cycles.

References

[1] Leonard D. Baumert and Daniel M. Gordon, *On the existence of cyclic difference sets with small parameters*, Fields Inst. Commun. **41** (2004), 29–40.

[2] Daniel M. Gordon, *The La Jolla Difference Set Repository*, ArasuFest talk slides, August 3, 2019.